# Notes: Babina, Fedyk, He, and Hodson (JFE, 2024)

**Artificial intelligence, firm growth, and product innovation**

## Contents

# 1 Big picture

- **Question.** Do firm investments in artificial intelligence generate economically meaningful growth, and if so, through which channel?
- **Measurement problem.** There is no standard firm-level dataset on AI adoption. The paper solves this by building human-capital-based measures of AI using two independent labor-market datasets: employee resumes and job postings.
- **Main empirical setting.** U.S. public firms from 2010 to 2018. The paper focuses on non-tech firms in the main regressions because it wants to study the *use* of AI across the economy rather than the invention of AI inside tech firms.
- **Main descriptive fact.** AI investments rise dramatically over the 2010s and spread across many sectors, not just the information sector.
- **Main result.** Firms that invest more in AI grow faster in sales, employment, and market value. In the baseline long-differences results, a one-standard-deviation increase in AI investments is associated with about 19.5% higher sales growth, 18.1% higher employment growth, and 22.3% higher growth in market value from 2010 to 2018.
- **Mechanism result.** The main channel is product innovation, not cost cutting. AI-investing firms file more trademarks, produce more product patents, and change their product mix more, but they do *not* show robust gains in sales per worker, revenue TFP, or process patents.
- **Identification strategy.** The paper combines rich controls, dynamic lead-lag tests, and an IV based on firms' pre-2010 exposure to universities that were historically strong in AI research.
- **Heterogeneity result.** The AI-growth relationship is much stronger for initially larger firms, consistent with AI favoring firms that already have more scale and more data.
- **Industry result.** Industries that invest more in AI see higher sales and employment among Compustat firms, but also greater concentration.
- **Economic takeaway.** In this paper, AI looks less like a pure labor-saving automation shock and more like a technology that helps firms create and improve products. That is why it is associated with growth and with stronger superstar-firm dynamics.

# 2 Data and measurement (key objects)

## 2.1 Main data sources

- **Cognism resumes.** Roughly 535 million individual profiles globally. In the U.S. data from 2007 to 2018, the authors observe 657 million person-firm-year employment records, of which 120 million are matched to U.S. public firms. The matched sample contains 19 million distinct employees.
- **Burning Glass job postings.** Over 180 million U.S. job postings in 2007 and 2010–2018, with rich tags for required skills.

- **Compustat.** Sales, employment, market value, balance-sheet variables, firm age, and other operating outcomes.
- **Supplementary data.** ACS and QWI for local labor markets and wages; Open Academic Graph and HERDS for university AI research; USPTO trademarks and patents; Hoberg-Phillips text-based product descriptions for product-mix changes.

*Interpretation.* The paper's empirical contribution starts with measurement. Without the labor-market data, there is no plausible way to see actual firm-level AI adoption outside a few surveys or case studies.

## 2.2 AI investments from job postings

The authors first learn which *skills* are AI-related from the job-postings data. For each skill $s$, they compute

$$w_s^{AI} = \frac{\#\{\text{jobs requiring skill } s \text{ and also mentioning AI, ML, NLP, or CV}\}}{\#\{\text{jobs requiring skill } s\}}. \tag{1}$$

Here the core AI terms are artificial intelligence, machine learning, natural language processing, and computer vision.

Then the AI-relatedness of a posting $j$ is the average of its required-skill scores:

$$\omega_j^{AI} = \frac{1}{|S_j|} \sum_{s \in S_j} w_s^{AI}. \tag{2}$$

A posting is classified as AI-related if $\omega_j^{AI} > 0.1$. The firm-level job-posting measure is therefore

$$ShareAI_{f,t}^{post} = \frac{\#\{\text{AI-related postings by firm } f \text{ in year } t\}}{\#\{\text{all postings by firm } f \text{ in year } t\}}. \tag{3}$$

*Interpretation.* This is a data-driven classifier. Instead of manually choosing a small keyword list, the paper lets the skill co-occurrence patterns reveal which skills are genuinely specific to AI. That sharply reduces the risk of labeling generic programming or statistics work as AI.

## 2.3 AI investments from resumes

The paper's main measure comes from actual employees rather than vacancies. Starting from the 67 most AI-related keywords learned from job postings, the authors classify an employee-firm-year as AI-related if any of the top AI terms appears in the worker's:

- job title,
- job description,
- patents obtained during that job spell (allowing for the patent lag),
- publications, or
- awards.

The baseline firm-level measure is

$$ShareAI^{resume}_{f,t} = \frac{\#\{\text{AI-classified employees at firm } f \text{ in year } t\}}{\#\{\text{all employees at firm } f \text{ in year } t\}}. \tag{4}$$

*Interpretation.* This is the paper's preferred measure because it captures *actual hiring and actual employee stocks.* If a firm posts AI jobs but cannot fill them, job postings overstate adoption. Resume data are also more likely to capture AI talent obtained through acquisitions.

## 2.4 Sample and basic coverage

- Of 112 million job postings with non-missing employer names and required skills, 42 million are matched to Compustat firms.
- Publicly listed firms account for about 38% of all job postings in the matched Burning Glass sample, but about half of all AI-related postings, suggesting that public firms are relatively intensive users of AI talent.
- The Cognism data slightly over-represent high-skilled workers, but still cover about 64% of the U.S. workforce as of 2018.
- The main firm-level regressions exclude two 2-digit NAICS sectors: Information and Professional, Scientific, and Technical Services.

*Interpretation.* Excluding tech sectors is not a robustness check; it is part of the paper's design. The authors want to know whether AI works like a broad, general-purpose technology outside the firms that build the core technology itself.

## 2.5 Descriptive patterns that validate the measure

- The resume-based AI share rises from 0.04% in 2007 to 0.29% in 2018.
- The job-postings-based AI share rises from 0.10% in 2010 to 0.80% in 2018.
- The information sector has the highest level of AI adoption, but almost all sectors experience meaningful growth in AI jobs.
- Changes in the resume-based AI share have a correlation of about 0.27 with changes in log R&D expenditures from 2010 to 2018 (controlling for industry fixed effects).
- Across firms, the posting-based and resume-based AI measures are positively correlated. The raw Spearman correlation for counts rises from about 0.32 in 2010 to 0.56 in 2018; for AI-job shares it rises from about 0.27 to 0.47.

*Interpretation.* These facts matter because the measures are novel. The paper needs to show not just that AI rises, but that the rise occurs where we would expect it to occur and that two independent datasets tell a similar story.

## 2.6 Other important variables

The outcomes used later in the paper are built as follows.

- **Growth outcomes:** changes in log sales, log employment, and log market value from 2010 to 2018.
- **Product innovation:** changes in log(1 + #trademarks), log(1 + #product patents), and a text-based measure of changes in product mix from 10-K product descriptions.
- **Operating-cost / productivity outcomes:** changes in log cost of goods sold, log operating expenses, log sales per worker, and revenue TFP.
- **Process innovation:** changes in log(1 + #process patents).
- **Industry outcomes:** industry sales, industry employment, HHI, and the market share of the largest firm.

*Interpretation.* The key empirical distinction in the paper is between *product-creation outcomes* and *cost-reduction / productivity outcomes.* That is how the authors separate product innovation from process innovation.

# 3 Models and empirical specifications (all equations + interpretation)

## 3.1 Conceptual mechanism: product innovation versus process innovation

The paper organizes the economics around two non-mutually-exclusive channels:

- **Product innovation channel.** AI lowers the cost of experimenting with and developing new products, improves existing products, and helps firms tailor products to customer tastes.
- **Process innovation channel.** AI lowers operating costs, automates tasks, improves forecasting, and raises productivity for existing products.

*Interpretation.* If the first channel dominates, we should see more trademarks, more product patents, and more change in product offerings. If the second dominates, we should see lower operating costs, higher productivity, and more process patents. The whole mechanism section is built around this contrast.

## 3.2 Baseline long-differences regression

The main regression is

$$\Delta Y_{i,[2010,2018]} = \beta\, \Delta ShareAI_{i,[2010,2018]} + Controls'_{i,2010}\gamma + SectorFE_i + \varepsilon_i. \tag{5}$$

Here:

- $Y$ is log sales, log employment, or log market value,
- $\Delta ShareAI$ is the change in the resume-based AI share from 2010 to 2018,
- the AI variable is standardized to mean zero and standard deviation one,

- controls are measured in 2010,
- the regressions are weighted by the number of Cognism resumes in 2010,
- standard errors are clustered at the 5-digit NAICS level.

The baseline controls include:

- firm-level characteristics: log sales, cash/assets, firm age, and log number of resumes,
- log industry wage,
- commuting-zone characteristics: the shares of IT workers, college-educated workers, foreign-born workers, routine workers, finance/manufacturing workers, female workers, and local average wage.

*Interpretation.* This is a classic long-differences design for slow-moving technology adoption. By differencing 2010 to 2018, the paper removes time-invariant firm heterogeneity and focuses on medium-run adjustment rather than short-run noise.

## 3.3 Main growth coefficients

The headline results in Table 3 are:

$$\beta_{sales} = 0.195, \qquad \beta_{employment} = 0.181, \qquad \beta_{mktval} = 0.223, \tag{6}$$

with all three estimated using the fully controlled specification.

*Interpretation.* Because the AI variable is standardized, these coefficients mean that a one-standard-deviation increase in AI investment intensity from 2010 to 2018 is associated with about 19.5% more sales growth, 18.1% more employment growth, and 22.3% more market-value growth over the same period.

## 3.4 Dynamic lead-lag specification

To test pre-trends and timing, the paper estimates a distributed lead-lag model:

$$Y_{it} = \sum_{k=-2}^{5} \delta_k \, \Delta ShareAI_{i,t-k} + \mu_i + \lambda_{nt} + \theta_{st} + \varepsilon_{it}, \tag{7}$$

where $\mu_i$ are firm fixed effects, $\lambda_{nt}$ are 2-digit-NAICS-by-year fixed effects, and $\theta_{st}$ are state-by-year fixed effects.

*Interpretation.* The lead coefficients test whether AI-investing firms were already on different growth paths before investment. The lag coefficients show how long it takes for the gains from AI to show up.

Empirically, the paper finds:

- no evidence of pre-trends,
- sales and employment rise with a lag of roughly two to three years,
- the cumulative response is around 1.5%–2% in annual sales growth and similar for employment.

## 3.5 Heterogeneity by initial firm size

To study whether AI benefits large firms more, the paper interacts AI growth with terciles of initial firm size (measured within 2-digit NAICS sectors using 2010 employment).

*Interpretation.* This is a direct test of the "big data favors big firms" idea. If large firms have more data and better complementary assets, AI should be more powerful for them.

With controls included, the coefficients for sales growth are approximately:

$$\beta_{small} \approx -0.001, \qquad \beta_{middle} \approx 0.183, \qquad \beta_{large} \approx 0.204. \tag{8}$$

For employment growth they are about −0.021, 0.157, and 0.182, and for market value about 0.008, 0.170, and 0.242.

*Interpretation.* Once controls are added, the smallest firms show essentially no measurable growth effect, while the medium and large firms do. This is strong evidence that the gains from AI are concentrated higher up the size distribution.

## 3.6 Mechanism regressions

The paper reuses the long-differences design with different dependent variables:

$$\Delta ProductInnovation_i = \beta\, \Delta ShareAI_i + Controls_i + SectorFE_i + \varepsilon_i, \tag{9}$$

$$\Delta Productivity_i = \beta\, \Delta ShareAI_i + Controls_i + SectorFE_i + \varepsilon_i, \tag{10}$$

$$\Delta ProcessInnovation_i = \beta\, \Delta ShareAI_i + Controls_i + SectorFE_i + \varepsilon_i. \tag{11}$$

*Interpretation.* The identification logic is the same as in the growth regressions. Only the outcomes change. That makes the comparison between the product and process channels very clean.

## 3.7 Industry-level regression

To study aggregate implications, the paper aggregates AI exposure to the 5-digit NAICS industry level and estimates

$$\Delta \ln y_{j,[2010,2018]} = \gamma\, \Delta ShareAI_{j,[2010,2018]} + Controls'_{j,2010}\eta + SectorFE_j + \varepsilon_j, \tag{12}$$

where $y_j$ is industry sales, employment, HHI, or the top-firm market share.

*Interpretation.* This asks whether AI mainly reshuffles business across firms or whether it also shows up as higher growth at the industry level. The concentration outcomes then ask whether the gains are tilted toward the biggest firms.

# 4 Identification and instruments (what makes the estimates credible)

## 4.1 Main endogeneity concerns

There are three obvious threats to causal interpretation.

- **Reverse causality.** Firms that were already about to grow may have invested more in AI.
- **Omitted technology shocks.** AI investment may proxy for broader IT, robotics, or data-analytics investment.
- **Demand shocks.** Positive industry or local demand shocks could simultaneously raise growth and AI hiring.

The paper addresses these in layers:

- rich 2010 firm, industry, and commuting-zone controls,
- controls for pre-2010 firm and industry growth and for Tobin's Q,
- dynamic lead-lag tests showing no pre-trends,
- direct controls for non-AI IT, robotics, non-AI data skills, and non-AI data analytics,
- an instrumental-variables design based on AI labor supply.

*Interpretation.* No single test is decisive on its own. The paper's credibility comes from the fact that all of these checks point in the same direction.

## 4.2 How the instrument is built

The core idea is that AI talent is scarce, and firms connected *before 2010* to universities that were historically strong in AI should have had a differential ability to hire AI workers once commercial interest in AI took off after 2012.

A simplified version of the instrument is

$$Z_i = \sum_u s_{iu}^{2010} \, AIstrong_u, \tag{13}$$

where:

- $s_{iu}^{2010}$ is firm $i$'s pre-2010 hiring-network weight on university $u$, measured from employees' alma maters in 2010 (with special attention to STEM workers),
- $AIstrong_u$ indicates that university $u$ was historically strong in AI research before 2010.

The paper also constructs analogous exposure controls to universities strong in broader computer science and to top-ranked universities.

*Interpretation.* The exclusion restriction is that pre-2010 links to AI-strong universities matter for post-2010 growth only because they relax firms' access to AI talent, not because they proxy for some other growth driver.

## 4.3 How AI-strong universities are identified

Using Open Academic Graph data and HERDS, the paper classifies researchers by publication field and then labels a university as AI-strong if, in at least one year from 2005 to 2009, either:

- its number of AI researchers is in the top 5% across universities, or

- its number of AI researchers is in the top 10% and its share of AI researchers is in the top 5%.

The authors show that their publication-based researcher counts are sensible: the cross-sectional correlation between the log number of university researchers in OAG and log R&D spending in HERDS is about 0.83.

*Interpretation.* This part matters because the IV hinges on a pre-existing supply shock. The paper wants universities that were strong in AI *for research reasons* before firms broadly wanted AI workers.

## 4.4 Why the first stage is plausible

The resume data allow the authors to check whether AI-strong universities really produced more AI-trained graduates in the 2010s.

- In 2006 and even in 2012, the share of AI-trained fresh graduates is very low for both groups of universities.
- From 2012 to 2018, the share roughly triples to about 1.5% at AI-strong universities, while remaining below 0.5% at non-AI-strong universities.
- The resume data cover about 59% of all degrees conferred by each university on average, and the number of fresh graduates in the resume data is highly correlated with IPEDS graduation counts (correlation about 0.73).

*Interpretation.* This is the economic engine of the instrument: historically AI-strong universities differentially increase the supply of AI-ready graduates exactly when the commercial demand for AI talent surges.

## 4.5 Why the instrument is not obviously invalid

The paper spends a lot of effort on exclusion-restriction concerns.

- AI-strong universities are not just the same as top-ranked universities.
- The regressions control for exposure to non-AI CS-strong universities and top-10 universities.
- Firms connected to AI-strong universities in 2010 did *not* increase those connections from 2005 to 2010 in anticipation of later AI demand.
- Firms more exposed to AI-strong universities did *not* grow faster before 2010.
- The first-stage F-statistics are above 10 in all specifications and near 20 in the richer ones.

*Interpretation.* This is a good instrument for the margin the paper cares about: relaxing the supply constraint for AI labor. It is still not a perfect randomized shock, but the design is unusually thoughtful for a technology-adoption paper.

## 4.6 IV results

Table 5 shows that the 2SLS estimates are, if anything, larger than OLS. In the richest specification:

$$\beta^{IV}_{sales} = 0.317, \qquad \beta^{IV}_{employment} = 0.330, \qquad \beta^{IV}_{mktval} = 0.360. \tag{14}$$

Across columns the IV coefficients range roughly from 0.32 to 0.71, and the first-stage F-statistics range from about 12.7 to 19.5.

*Interpretation.* The main message is not that IV is dramatically larger than OLS, but that the positive growth effect survives a very different source of identifying variation. That is what makes the case for a causal interpretation much stronger.

## 4.7 What the paper still cannot fully rule out

- The IV identifies the effect of easier access to AI talent, not every possible dimension of AI adoption.
- Some residual concern always remains that historically AI-strong universities proxy for unobserved firm quality or network quality.
- The analysis is mostly about publicly listed firms, so it may not generalize one-for-one to small private firms.

*Interpretation.* These limitations do not overturn the results, but they help frame the exact scope of the paper's conclusions.

# 5 Tables and figures

## 5.1 Figure 1 and Figure 2: AI adoption rises fast and spreads broadly

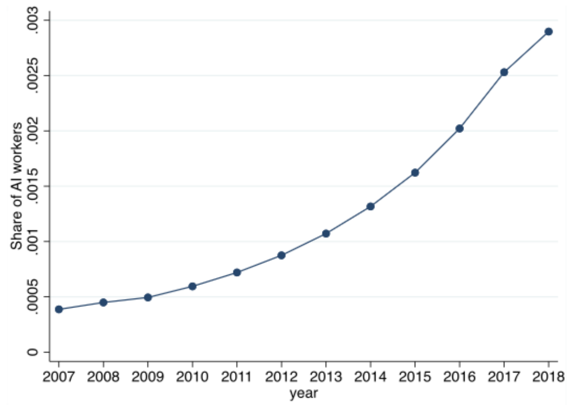
**Read:**

- Figure 1 shows that AI-related labor rises sharply over time in both datasets. The resume-based AI share rises from 0.04% in 2007 to 0.29% in 2018; the job-postings-based share rises from 0.10% in 2010 to 0.80% in 2018.
- Figure 2 shows that the information sector has the highest level of AI adoption, but the increase is visible across many sectors.

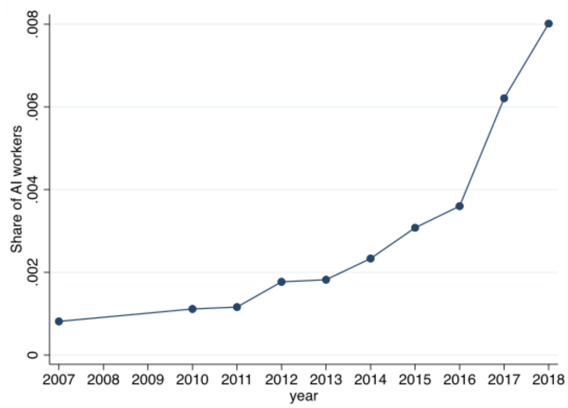
**Economic message.** These figures establish the paper's two basic stylized facts: AI adoption is economically meaningful even though AI workers are rare, and the technology is not confined to a narrow set of firms.

## 5.2 Table 1 and Table 2: the new AI measures behave sensibly

**Read:**

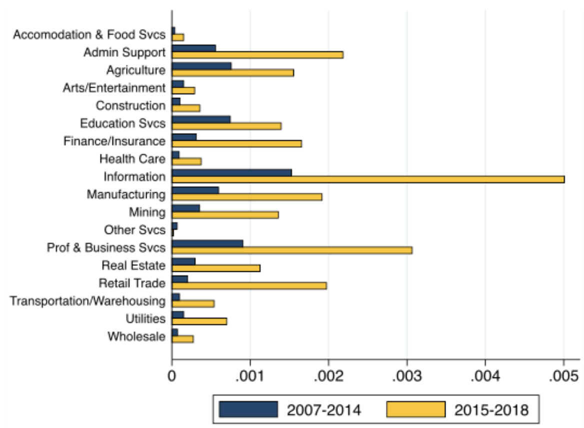



(a) Resume data (Cognism)

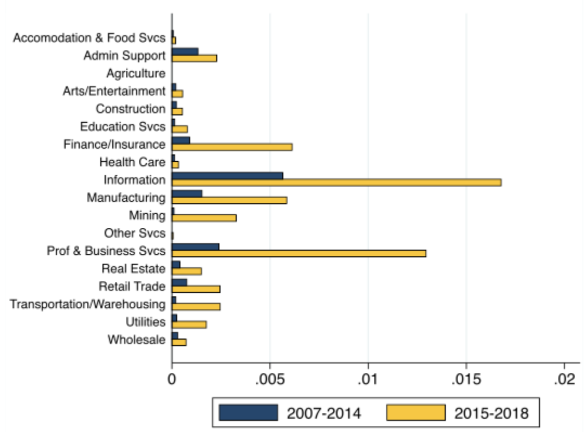



(b) Job posting data (Burning Glass)

**Fig. 1.** Time Series of AI Investments. This figure shows the time series of the two measures of AI investments for U.S. public firms. Panel (a) shows the fraction of all employees in a given year who are classified as holding AI-related positions in the Cognism resume data from 2007 to 2018. Panel (b) reports the fraction of job postings in the Burning Glass data (with job-level continuous AI measure above 0.1) for 2007 and 2010–2018.




(a) Resume data (Cognism)




(b) Job posting data (Burning Glass)

**Fig. 2.** AI Investments by Industry Sector. This figure presents the average share of AI jobs at the industry level, based on the sample of U.S. public firms. For each sector (based on 2-digit NAICS industry codes), we compute the fraction of AI-related employees in the Cognism resume data (in Panel (a)) and the average share of AI-related job postings in the Burning Glass data (with job-level continuous AI measure above 0.1) across all job postings (in Panel (b)). The statistics are computed for firms in each industry sector across two sub-periods: 2007–2014 and 2015–2018. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

**Table 1**
Correlations between Job-posting-based and Resume-based AI Measures. This table reports, for each year from 2010 to 2018, the Spearman rank correlations between three pairs of firm-level variables: (i) the absolute number of AI job postings in Burning Glass against the absolute number of AI employees in resumes from Cognism; (ii) the fraction of employees classified as AI-related in the two datasets; and (iii) the fraction of AI employees in Cognism against the average continuous AI measure in Burning Glass. Panel 1 shows raw correlations, and Panel 2 displays correlations conditional on the 2-digit NAICS industry sector fixed effects and the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes and job postings), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). All correlations are computed over the cross-section of firms with at least 20 U.S. jobs in the Cognism resume data in each year of the sample.

**Panel 1: Raw Correlations**

| Year | Correlations between: Numbers of AI jobs | Fractions of AI Jobs | Cognism fraction & BG continuous measure |
|---|---|---|---|
| 2010 | 0.320 | 0.272 | 0.374 |
| 2011 | 0.341 | 0.288 | 0.390 |
| 2012 | 0.338 | 0.291 | 0.388 |
| 2013 | 0.424 | 0.363 | 0.447 |
| 2014 | 0.468 | 0.410 | 0.484 |
| 2015 | 0.474 | 0.405 | 0.496 |
| 2016 | 0.503 | 0.421 | 0.499 |
| 2017 | 0.564 | 0.474 | 0.531 |
| 2018 | 0.574 | 0.484 | 0.538 |

**Panel 2: Correlations Conditional on Baseline Controls**

| Year | Correlations between: Numbers of AI jobs | Fractions of AI Jobs | Cognism fraction & BG continuous measure |
|---|---|---|---|
| 2010 | 0.825 | 0.650 | 0.470 |
| 2011 | 0.822 | 0.622 | 0.476 |
| 2012 | 0.801 | 0.583 | 0.487 |
| 2013 | 0.784 | 0.569 | 0.498 |
| 2014 | 0.757 | 0.526 | 0.551 |
| 2015 | 0.729 | 0.467 | 0.513 |
| 2016 | 0.702 | 0.475 | 0.501 |
| 2017 | 0.687 | 0.507 | 0.510 |
| 2018 | 0.670 | 0.502 | 0.513 |

**Table 2**
Firm-level Determinants of Resume-based AI Investments. This table reports the coefficients from regressions of cross-sectional changes in AI investments by U.S. public firms (in non-tech sectors) from 2010 to 2018 on the following ex-ante firm characteristics measured in 2010: log sales in column 1, cash/assets in column 2, R&D/sales in column 3, revenue TFP in column 4, log markup measured following De Loecker et al. (2020) in column 5, Tobin's Q in column 6, market leverage in column 7, return on assets (ROA) in column 8, and firm age in column 9. The dependent variable is the growth in the share of AI workers from 2010 to 2018 using the resume data from Cognism. Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for the 2-digit NAICS industry sector fixed effects. The dependent variable is normalized to have a mean of zero and a standard deviation of one. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

| | Δ Share of AI Workers, 2010–2018 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
| Log Sales 2010 | 0.109*** | | | | | | | | | 0.175*** |
| | (0.038) | | | | | | | | | (0.033) |
| Cash/Assets 2010 | | 3.986*** | | | | | | | | 3.233*** |
| | | (1.242) | | | | | | | | (0.832) |
| R&D/Sales 2010 | | | 3.734 | | | | | | | 1.062 |
| | | | (2.313) | | | | | | | (1.745) |
| Revenue TFP 2010 | | | | 1.386 | | | | | | 0.567 |
| | | | | (1.038) | | | | | | (0.532) |
| Log Markup 2010 | | | | | 0.418 | | | | | 0.147 |
| | | | | | (0.312) | | | | | (0.177) |
| Tobin's Q 2010 | | | | | | 0.335** | | | | 0.158 |
| | | | | | | (0.153) | | | | (0.118) |
| Market Leverage 2010 | | | | | | | -0.973 | | | 0.213 |
| | | | | | | | (0.690) | | | (0.301) |
| ROA 2010 | | | | | | | | 1.409** | | 0.164 |
| | | | | | | | | (0.656) | | (1.127) |
| Firm Age 2010 | | | | | | | | | -0.003 | -0.004** |
| | | | | | | | | | (0.004) | (0.002) |
| Industry FE | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| Adj R-Squared | 0.151 | 0.309 | 0.172 | 0.148 | 0.142 | 0.215 | 0.129 | 0.129 | 0.118 | 0.418 |
| Observations | 1,731 | 1,731 | 1,731 | 1,731 | 1,731 | 1,731 | 1,731 | 1,731 | 1,731 | 1,731 |

- Table 1 shows that the posting-based and resume-based measures are positively correlated, with the correlations rising over time.
- Table 2 shows that larger firms and cash-rich firms invest more in AI. In the multivariate specification, log sales and cash/assets robustly predict future AI adoption, while TFP, markups, Tobin's Q, leverage, and ROA do not do so consistently.

**Economic message.** Table 1 says the new measures are not random noise. Table 2 then tells an important economic story: AI adoption is easier for firms with scale and slack resources. That already hints that the benefits of AI may not be evenly distributed.

## 5.3 Table 3, Table 4, and Figure 3: AI predicts growth, especially for larger firms

**Read:**

- Table 3 is the paper's baseline result: a one-standard-deviation increase in the AI-worker share from 2010 to 2018 predicts about 19.5% more sales growth, 18.1% more employment growth, and 22.3% more market-value growth.
- Table 4 shows that these effects are much stronger for initially larger firms. Once controls are included, the smallest tercile has almost no sales-growth effect, while the middle and largest terciles have coefficients around 0.18 to 0.20.
- Figure 3 shows no pre-trends and a lagged dynamic response: growth starts to rise materially about two to three years after AI investment.

**Economic message.** This is the heart of the paper. AI is associated with expansion, not contraction, and the benefits arrive with a lag consistent with organizational learning and experimentation. The size heterogeneity also foreshadows the concentration results later on.

## 5.4 Table 5: the IV confirms the growth effect

**Read:**

**Table 3**

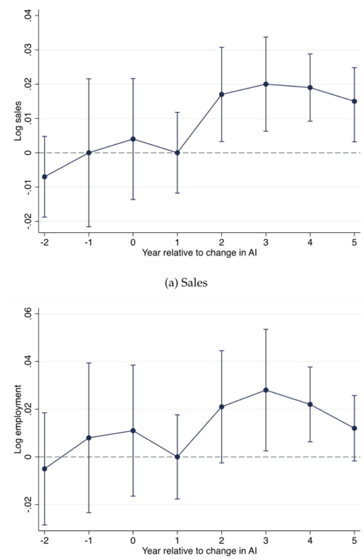
AI Investments and Firm Growth: Long-differences Estimates Using the Resume-based AI Measure. This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on the contemporaneous firm-level changes in AI investments among U.S. public firms (in non-tech sectors). We consider three measures of firm growth: changes in log sales (columns 1 and 2), changes in log employment (columns 3 and 4), and changes in log market value (columns 5 and 6). The main independent variable is growth in the share of AI workers (based on the resume data) from 2010 to 2018, standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for the 2-digit NAICS industry sector fixed effects. Columns 2, 4, and 6 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

| | Δ Log Sales | | Δ Log Employment | | Δ Log Market Value | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Δ Share AI Workers | 0.202*** | 0.195*** | 0.239** | 0.181** | 0.240** | 0.223** |
| | (0.069) | (0.069) | (0.097) | (0.086) | (0.093) | (0.086) |
| Industry FE | Y | Y | Y | Y | Y | Y |
| Controls | N | Y | N | Y | N | Y |
| Adj R-Squared | 0.221 | 0.422 | 0.237 | 0.405 | 0.247 | 0.364 |
| Observations | 1,052 | 1,052 | 1,052 | 1,052 | 1,009 | 1,009 |

**Table 4**

Heterogeneous Relationship between AI Investments and Firm Growth by Initial Firm Size Using the Resume-based AI Measure. This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on contemporaneous changes in AI investments among U.S. public firms (in non-tech sectors), separately for each tercile of initial firm size. Firms in each 2-digit NAICS industry sector are divided into terciles based on employment in 2010. We consider three measures of firm-level growth for the dependent variable: changes in log sales (columns 1 and 2), changes in log employment (columns 3 and 4), and changes in log market value (columns 5 and 6). The main independent variable is the growth in the share of AI workers (based on the resume data) from 2010 to 2018, standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for the 2-digit NAICS industry sector interacted with initial firm size tercile fixed effects. Columns 2, 4, and 6 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. We report the t-statistics and p-values from the t-tests for the difference between the coefficient of Δ*Share AI Workers*Size Tercile* 1 and the coefficient of Δ*Share AI Workers*Size Tercile 3* at the bottom of the table. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

| | Δ Log Sales | | Δ Log Employment | | Δ Log Market Value | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Δ Share AI Workers*Size Tercile 1 | 0.046** | -0.001 | 0.041** | -0.021 | 0.057* | 0.008 |
| | (0.023) | (0.021) | (0.018) | (0.031) | (0.030) | (0.051) |
| Δ Share AI Workers*Size Tercile 2 | 0.219*** | 0.183*** | 0.217*** | 0.157*** | 0.210*** | 0.170*** |
| | (0.054) | (0.050) | (0.048) | (0.055) | (0.045) | (0.050) |
| Δ Share AI Workers*Size Tercile 3 | 0.223*** | 0.204*** | 0.260** | 0.182** | 0.260** | 0.242** |
| | (0.077) | (0.076) | (0.105) | (0.091) | (0.102) | (0.099) |
| NAICS2*Size tercile FE | Y | Y | Y | Y | Y | Y |
| Controls | N | Y | N | Y | N | Y |
| Adj R-Squared | 0.248 | 0.419 | 0.253 | 0.414 | 0.240 | 0.347 |
| Observations | 1,044 | 1,044 | 1,044 | 1,044 | 1,002 | 1,002 |
| T-test statistic | 3.7 | 6.8 | 3.7 | 5.3 | 4.4 | 8.8 |
| T-test p value | 0.054 | 0.010 | 0.054 | 0.022 | 0.038 | 0.003 |




(a) Sales

(b) Employment

**Fig. 3.** AI Investments and Firm Growth Over Time. This figure plots the coefficients from the distributed lead-lag model. The dependent variable is annual log sales in Panel (a) and log employment in Panel (b). The independent variable is the annual change in the share of AI workers in the Cognism resume data, standardized to have a mean of zero and a standard deviation of one. Regressions include firm-level sales (or employment) observations between 2010 and 2016 and control for firm fixed effects, 2-digit NAICS industry-by-year fixed effects, and state-by-year fixed effects. Regressions are weighted by the number of workers in the Cognism resume data. The vertical bars indicate 95% confidence intervals. Standard errors are clustered at the 5-digit NAICS level.

**Table 5**

AI Investments and Firm Growth: IV Estimates Using the Resume-based AI Measure. This table estimates the relationship between AI investments and firm growth from 2010 to 2018 for U.S. public firms (in non-tech sectors), where firm AI investments are instrumented with ex-ante firm-level exposure to AI-skilled graduates from AI-strong universities (see the definition of the instrument in Section 5.3 and the details of instrument construction in Appendix A). The independent variable is the change in the share of AI workers from 2010 to 2018 based on the resume data. Regressions are weighted by the number of Cognism resumes in 2010. The independent variable and the instrument are standardized to mean zero and standard deviation of one. We consider changes in log sales in columns 1 to 4, log employment in columns 5 to 8, and log market value in columns 9 to 12. All specifications control for the 2-digit NAICS industry sector fixed effects and ex-ante exposure to universities that are strong in computer science research as well as top 10 universities. Columns 2–4, 6–8, and 10–12 also control for the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Columns 3–4, 7–8, and 11–12 additionally control for firm-level changes in log sales and log employment from 2000 to 2008. Columns 4, 8, and 12 add state fixed effects. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. The first-stage F-statistics of the instrument are reported for all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

| | Δ Log Sales | | | | Δ Log Employment | | | | Δ Log Market Value | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| Δ Share AI Workers | 0.360*** | 0.510*** | 0.467*** | 0.317** | 0.466*** | 0.712*** | 0.584*** | 0.330** | 0.449*** | 0.529*** | 0.470*** | 0.360* |
| | (0.101) | (0.125) | (0.137) | (0.152) | (0.154) | (0.224) | (0.194) | (0.163) | (0.129) | (0.160) | (0.172) | (0.184) |
| Industry FE | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| University Control | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| Baseline Controls | N | Y | Y | Y | N | Y | Y | Y | N | Y | Y | Y |
| Control Pre-trend | N | N | Y | Y | N | N | Y | Y | N | N | Y | Y |
| State FE | N | N | N | Y | N | N | N | Y | N | N | N | Y |
| F Statistic | 12.7 | 19.3 | 15.7 | 18.7 | 12.7 | 19.3 | 15.7 | 18.7 | 12.8 | 19.5 | 15.5 | 19.3 |
| Observations | 1,001 | 1,001 | 777 | 773 | 1,001 | 1,001 | 777 | 773 | 963 | 963 | 753 | 746 |

**Table 6**

AI Investments and Product Innovation Using the Resume-based AI Measure. This table reports the coefficients from long-differences regressions of the changes in measures of product innovation from 2010 to 2018 on the contemporaneous changes in AI investments by U.S. public firms (in non-tech sectors). The dependent variables are the change in log(1+number of trademarks) in columns 1 and 2; the change in log(1+number of product patents) in columns 3 and 4; and the change in the product mix in columns 5 and 6. Product patents are patents with over 50% of the claims being product claims, following the categorization in Ganglmair et al. (2021). The change in the product mix is measured as the sum of annual changes from 2010 to 2018, where each annual change is the angle between the two word vectors indicating firms' product offerings in that year and the previous year (the word vectors are constructed as in Hoberg et al. (2014)). The main independent variable is the resume-based measure of the growth in the share of AI workers from 2010 to 2018, which is standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for the 2-digit NAICS industry sector fixed effects. Columns 2, 4, and 6 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Columns 1 and 2 control for the log number of trademarks from 2009 to 2010, and columns 3 and 4 control for the log number of patents from 2005 to 2010. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

| | Δ Log Number of Trademarks | | Δ Log Number of Product Patents | | Change in Product Mix | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Δ Share AI Workers | 0.144** | 0.134* | 0.221*** | 0.239*** | 0.149*** | 0.059 |
| | (0.065) | (0.078) | (0.035) | (0.031) | (0.036) | (0.038) |
| Industry FE | Y | Y | Y | Y | Y | Y |
| Controls | N | Y | N | Y | N | Y |
| Observations | 550 | 550 | 619 | 619 | 958 | 958 |

- Table 5 instruments AI investment with firms' pre-2010 exposure to AI-strong universities.
- The first stage is strong, with F-statistics above 10 and near 20 in the richer specifications.
- In the richest specification, the IV coefficient is about 0.317 for sales, 0.330 for employment, and 0.360 for market value.

**Economic message.** The IV matters because it uses labor-supply-driven variation in AI adoption rather than simply comparing fast-growing firms to slow-growing firms. The fact that the estimates remain strongly positive is a major credibility boost.

## 5.5 Table 6, Table 7, and Table 8: product innovation yes, productivity no

**Table 7**

AI Investments and Product Innovation: IV Estimates Using the Resume-based AI Measure. This table estimates the relationship between AI investments and product innovation from 2010 to 2018 for U.S. public firms (in non-tech sectors), where firms' AI investments are instrumented with ex-ante firm-level exposure to AI-skilled graduates from AI-strong universities (see the definition of the instrument in Section 5.3 and the details of instrument construction in Appendix A). The independent variable is the change in the share of AI workers from 2010 to 2018 based on the resume data. The independent variable and the instrument are standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Cognism resumes in 2010. We consider the change in log(1+number of trademarks) in columns 1 to 4, the change in log(1+number of product patents) in columns 5 to 8, and the change in the product mix in columns 9 to 12. Product patents are patents with over 50% of the claims being product claims, following the categorization in Ganglmair et al. (2021). The change in the product mix is measured as the sum of annual changes from 2010 to 2018, where each annual change is the angle between the two word vectors indicating firms' product offerings in that year and the previous year, following Hoberg et al. (2014). All specifications control for the 2-digit NAICS industry sector fixed effects and ex-ante exposure to universities that are strong in computer science research as well as top 10 universities. Columns 2–4, 6–8, and 10–12 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Columns 3, 4, 7, 8, 11, and 12 additionally control for firm-level changes in log sales and log employment from 2000 to 2008. Columns 4, 8, and 12 add state fixed effects. Columns 1–4 control for the log number of trademarks from 2009 to 2010, and columns 5–8 control for the log number of patents from 2005 to 2010. Standard errors are clustered at the 5-digit NAICS industry level, and reported in parentheses. The first-stage F-statistics of the instrument are reported for all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

| | Δ Log Number of Trademarks | | | | Δ Log Number of Product Patents | | | | Change in Product Mix | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| Δ Share AI Workers | 0.223 | 0.519 | 0.640** | 0.648** | 0.242 | 0.260 | 0.481*** | 0.430* | 0.185 | 0.187 | 0.281 | 0.314 |
| | (0.217) | (0.347) | (0.321) | (0.278) | (0.168) | (0.201) | (0.175) | (0.238) | (0.286) | (0.315) | (0.322) | (0.325) |
| Industry FE | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| University Control | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| Baseline Controls | N | Y | Y | Y | N | Y | Y | Y | N | Y | Y | Y |
| Control Pre-trend | N | N | Y | Y | N | N | Y | Y | N | N | Y | Y |
| State FE | N | N | N | Y | N | N | N | Y | N | N | N | Y |
| F Statistic | 11.8 | 10.5 | 8.5 | 15.9 | 11.5 | 23.8 | 36.9 | 34.6 | 13.1 | 22.2 | 16.0 | 21.8 |
| Observations | 528 | 528 | 435 | 426 | 586 | 586 | 479 | 469 | 932 | 932 | 725 | 717 |

**Table 8**

AI Investments and Operating Costs Using the Resume-based AI Measure. This table reports the coefficients from long-differences regressions of changes in firm operating costs and firm productivity from 2010 to 2018 on contemporaneous changes in AI investments by U.S. public firms (in non-tech sectors). The main independent variable is the change in the share of AI workers (based on the resume data) from 2010 to 2018, standardized to mean zero and standard deviation of one. We look at two measures of operating costs: log COGS in columns 1 and 2 and log operating expenses in columns 3 and 4. We consider two measures of productivity: log sales per worker (columns 5–6) and revenue TFP (columns 7–8). Revenue TFP is the residual from regressing log revenue on log employment and log capital (constructed using the perpetual inventory method), with separate regressions for each industry sector. In columns 9 and 10, the dependent variable is the change in log(1+number of process patents), where process patents are patents with over 50% of the claims being process claims, following the categorization in Ganglmair et al. (2021). Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for the 2-digit NAICS industry sector fixed effects. Columns 2, 4, 6, and 8 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Columns 9 and 10 also control for the log number of patents before 2010. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

| | Δ Log COGS | | Δ Log Operating Expense | | Δ Log Sales per Worker | | Δ Revenue TFP | | Δ Log Number of Process Patents | |
|---|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
| Δ Share AI Workers | 0.195*** | 0.177*** | 0.206*** | 0.191*** | -0.036 | 0.013 | -0.049 | -0.003 | -0.010 | 0.014 |
| | (0.052) | (0.053) | (0.066) | (0.065) | (0.028) | (0.022) | (0.046) | (0.037) | (0.039) | (0.054) |
| Industry FE | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| Controls | N | Y | N | Y | N | Y | N | Y | N | Y |
| Adj R-Squared | 0.213 | 0.390 | 0.237 | 0.420 | 0.176 | 0.322 | 0.222 | 0.358 | 0.700 | 0.747 |
| Observations | 1,052 | 1,052 | 1,052 | 1,052 | 1,052 | 1,052 | 977 | 977 | 619 | 619 |

**Read:**

- Table 6 shows that AI investments predict more trademarks and more product patents. With controls, a one-standard-deviation increase in AI is associated with about 13% more trademarks and about 23% more product patents.
- Table 7 shows that the IV estimates broadly reinforce this product-innovation channel, especially for trademarks and product patents, although the product-mix estimates are less precise.
- Table 8 shows no evidence that AI lowers costs or raises productivity. Cost of goods sold and operating expenses rise roughly one-for-one with expansion, while sales per worker, revenue TFP, and process patents do not increase significantly.

**Economic message.** This is the paper's sharpest mechanism result. AI helps firms *grow by making and improving products*, not by obviously making existing operations cheaper or more productive in the short to medium run.

## 5.6 Table 9: AI is linked to industry growth and concentration

**Read:**

- Table 9 shows that industries with greater AI-investment growth also experience greater sales and employment growth among Compustat firms.
- With controls, a one-standard-deviation increase in industry AI exposure is associated with about 0.173 higher log sales growth and 0.201 higher log employment growth.

**Table 9**
AI Investments and Changes in Industry Growth and Concentration Using the Resume-based AI Measure. This table reports the coefficients from industry-level long-differences regressions of the changes in industry sales, employment, and concentration on contemporaneous changes in industry-level AI investments. All industry-level variables are calculated for all firms in Compustat (regardless of whether they are in our main regression sample in Table 3 or not). Each observation is a 5-digit NAICS industry, and (as in our main analysis) we exclude tech sectors. The independent variable is the change in the share of AI workers (based on the resume data) from 2010 to 2018, standardized to mean zero and standard deviation of one. Regressions are weighted by the total (industry-level) number of Cognism resumes. The dependent variables are the changes, from 2010 to 2018, in log total sales in columns 1 and 2, log total employment in columns 3 and 4, the Herfindahl-Hirschman Index (HHI) in columns 5 and 6, and the market share of the top firm in an industry in columns 7 and 8. All specifications control for the 2-digit NAICS industry sector fixed effects. Regressions in columns 2, 4, 6, and 8 also include industry-level controls for log total employment, log total sales, and log average wage in 2010. Standard errors are robust against heteroskedasticity and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

| | Δ Log Sales | | Δ Log Employment | | Δ HHI | | Δ Top Firm Market Share | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Δ Share AI Workers | 0.169*** | 0.173*** | 0.195*** | 0.201*** | 0.018*** | 0.012* | 0.022*** | 0.014* |
| | (0.055) | (0.050) | (0.069) | (0.067) | (0.007) | (0.007) | (0.007) | (0.007) |
| Industry Sector FE | Y | Y | Y | Y | Y | Y | Y | Y |
| Controls | N | Y | N | Y | N | Y | N | Y |
| Observations | 275 | 275 | 275 | 275 | 275 | 275 | 275 | 275 |

- The same industries also become more concentrated: HHI and the market share of the largest firm both rise.

**Economic message.** This is where the firm-level results connect to the superstar-firm debate. AI does not just help adopters grow; it seems to help the biggest or best-positioned firms pull further ahead inside industries.

# 6 Short summary

- The paper builds new firm-level AI measures from employee resumes and job postings.
- AI adoption rises dramatically from 2010 to 2018 and spreads across sectors.
- Firms that invest more in AI grow faster in sales, employment, and market value.
- Dynamic and IV evidence both support a causal interpretation.
- The main mechanism is product innovation, not lower costs or higher productivity.
- The gains from AI are much stronger for larger firms.
- Industry AI adoption is associated with both more growth and more concentration.

# 7 Conclusion

## 7.1 Core conclusion

The paper's central conclusion is that AI has so far looked more like a product-innovation technology than a pure automation technology. Firms that invest in AI expand, innovate more, and become more valuable, but they do not obviously become more productive in the standard revenue-based sense.

## 7.2 Economic intuition

AI is a prediction technology. That makes it especially useful in settings where firms need to learn from data, experiment, match products to customer needs, and redesign offerings. Those are exactly the settings in which product innovation should accelerate. By contrast, if AI were mainly lowering the cost of existing production, we would expect to see cleaner gains in productivity and process innovation. The paper does not find that.

## 7.3 Why the paper matters

This paper matters for two reasons. First, it gives the literature a workable firm-level measurement strategy for AI adoption. Second, it changes the economic interpretation of AI's near-term effects. The evidence here suggests that AI can contribute to growth, but also to larger-firm dominance and higher industry concentration.

## 7.4 Takeaway

A compact way to remember the paper is: *AI helps firms grow mainly by helping them build and improve products, and the firms that benefit the most are the firms already positioned to scale.*